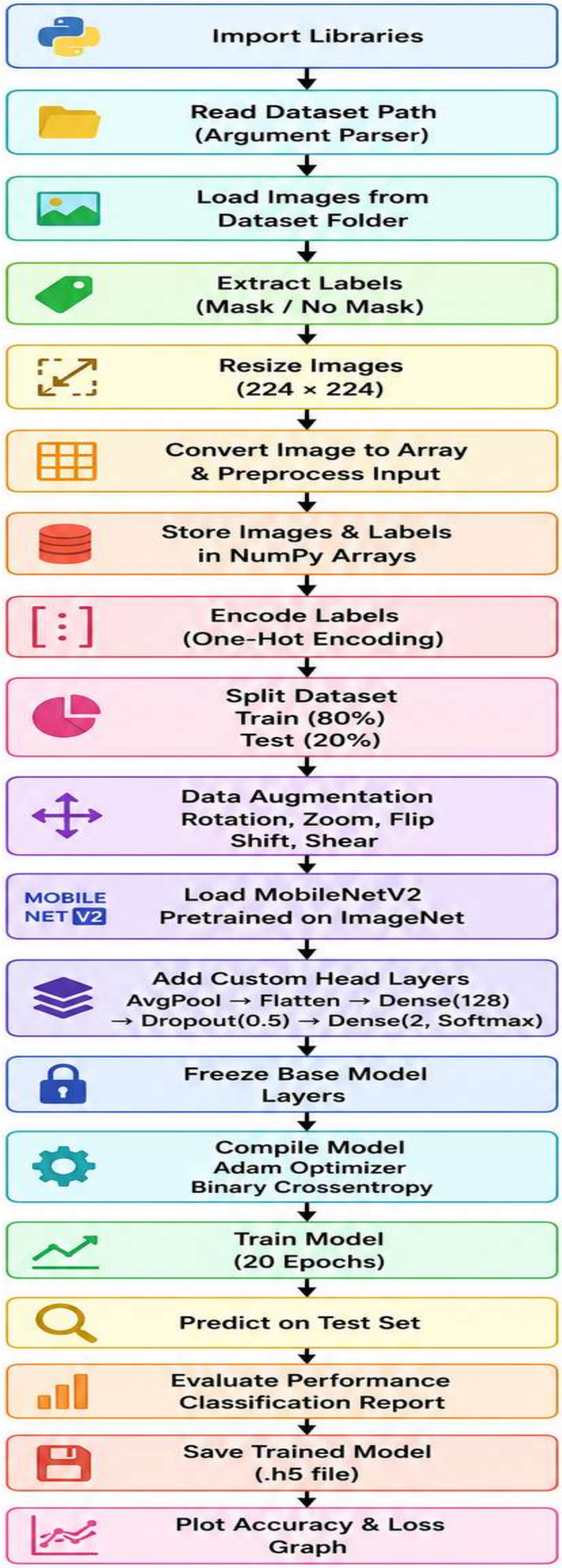
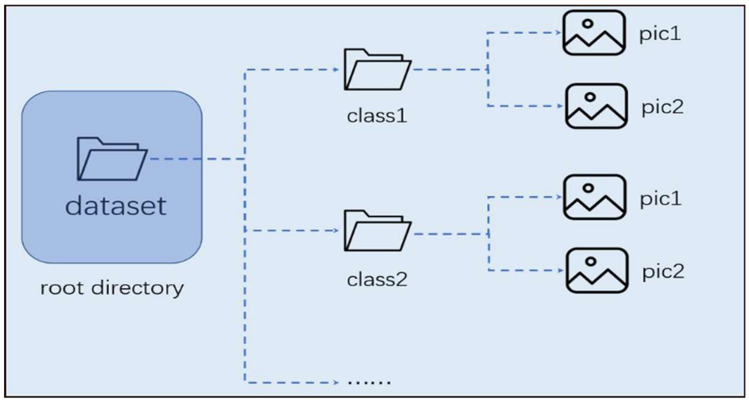
Detailed technical architecture



Import Libraries

Read Dataset Path



Load Images from Dataset Folder

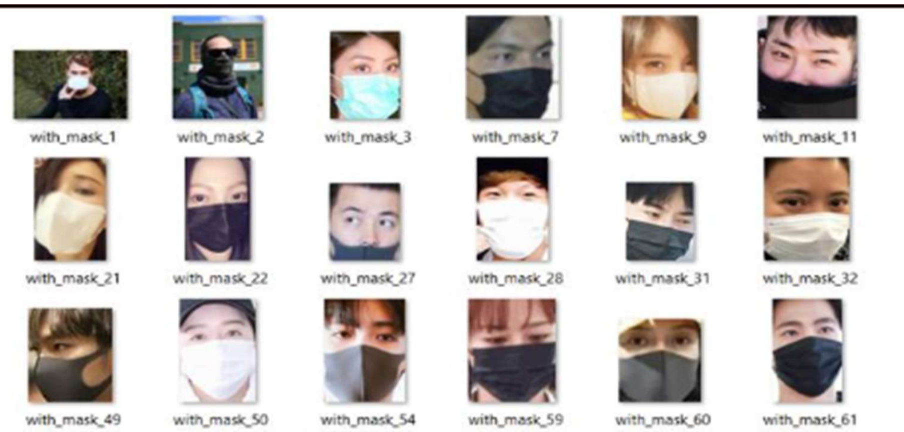
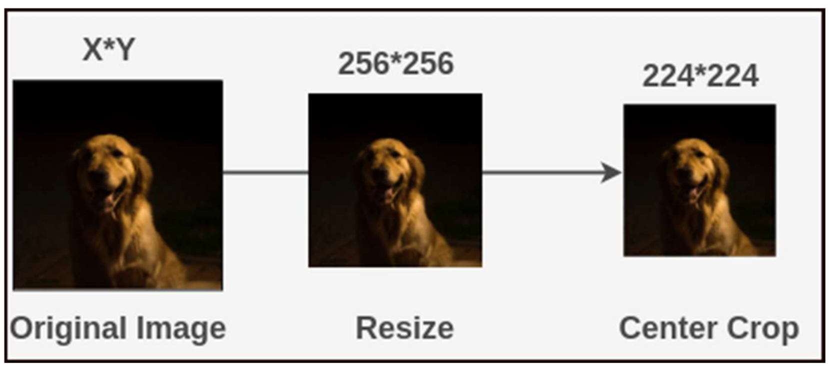


Fig 3: Collection (with mask)

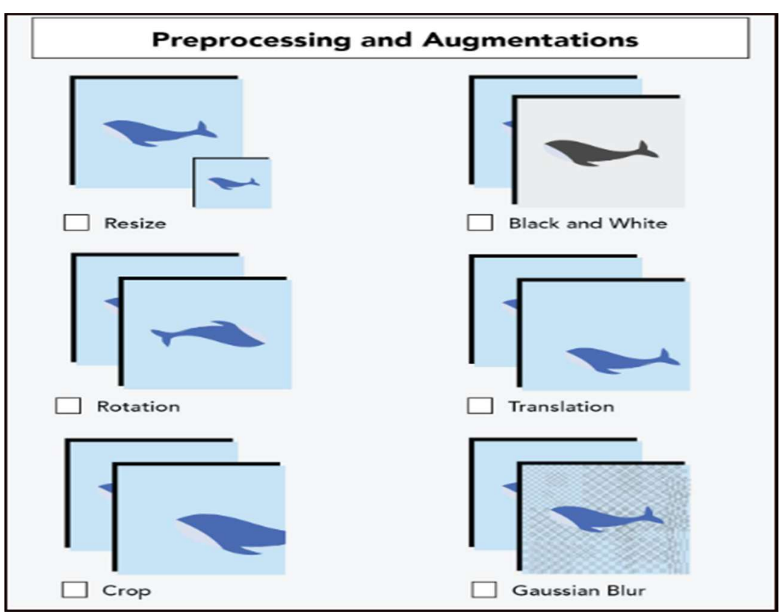
Fig 4: Collection (without mask)

Extract Labels (Mask / No Mask)

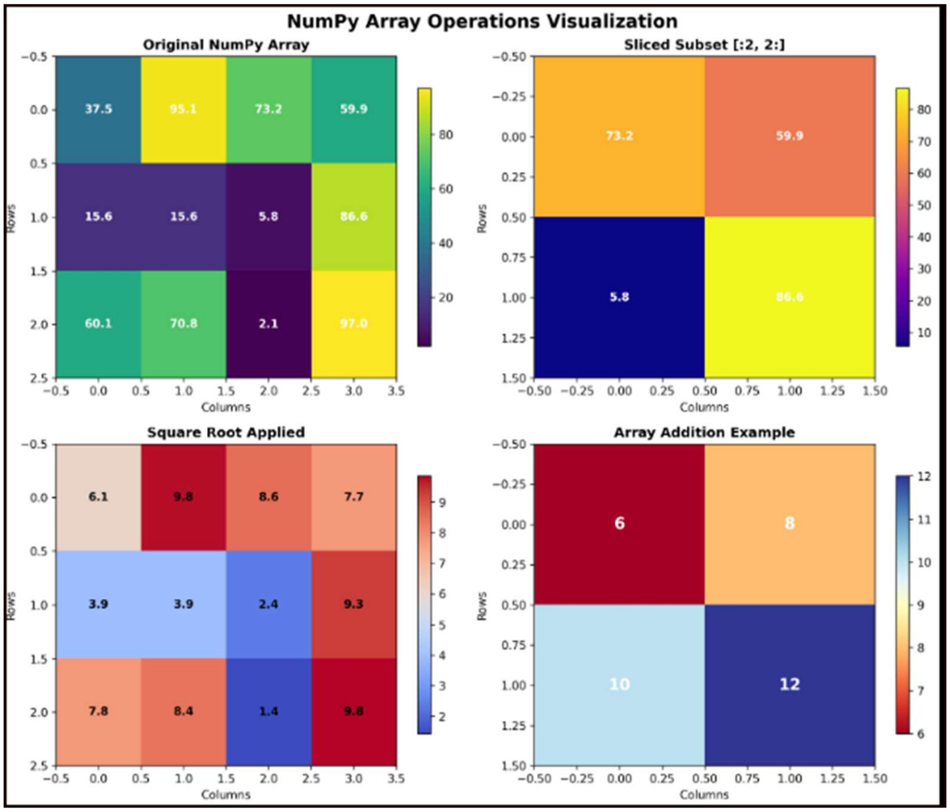
Resize Images (224×224)



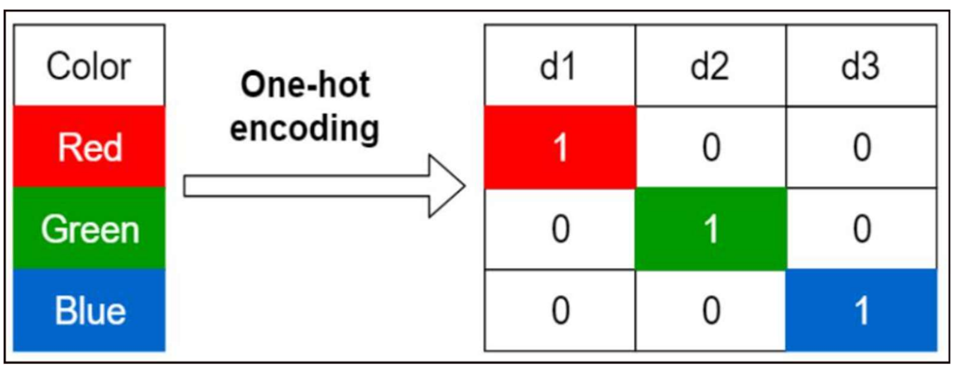
Convert Image to Array & Preprocess



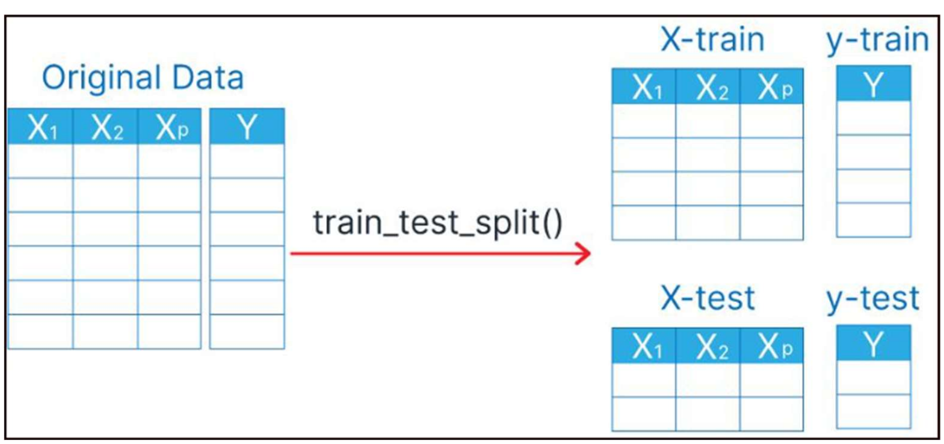
Store Images & Labels in NumPy Arrays



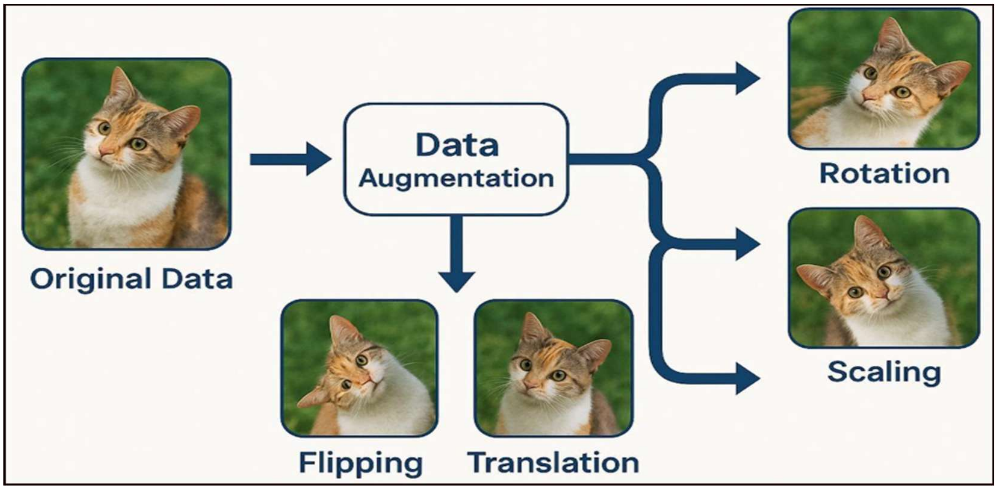
Encode Labels (One-Hot Encoding)



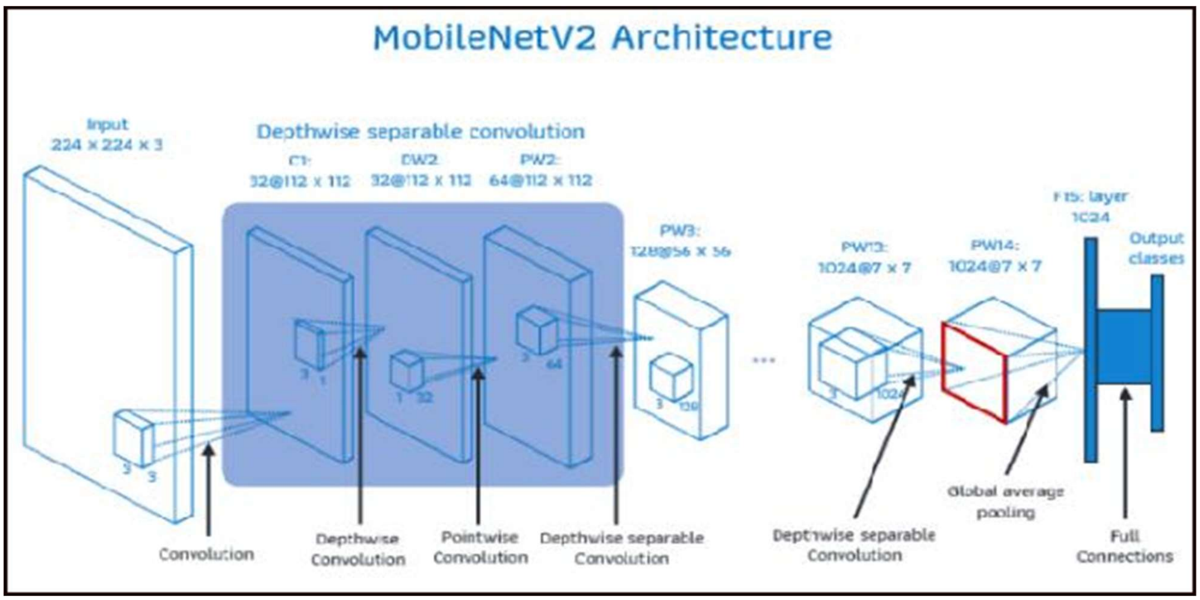
Split Dataset

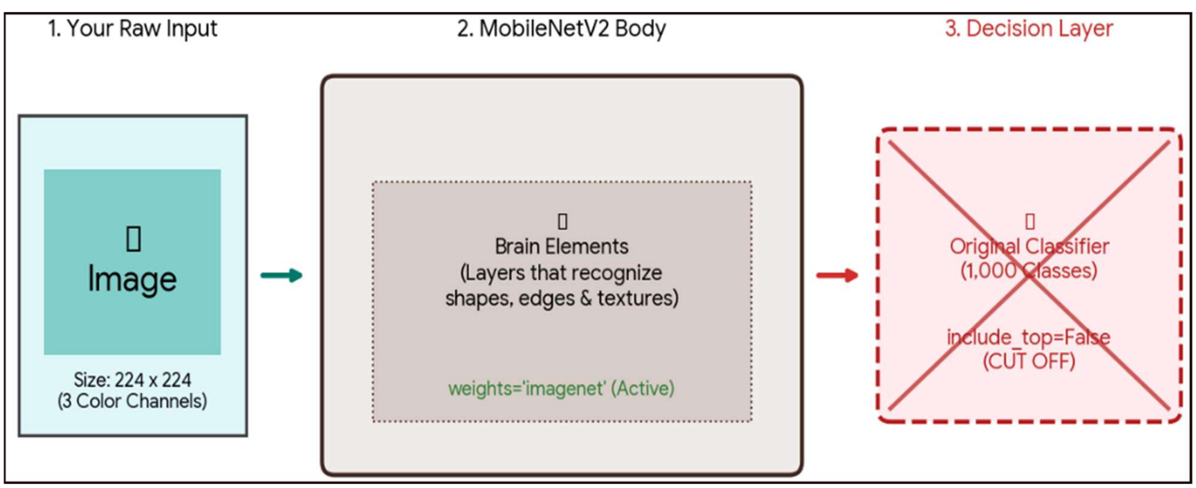


Data Augmentation

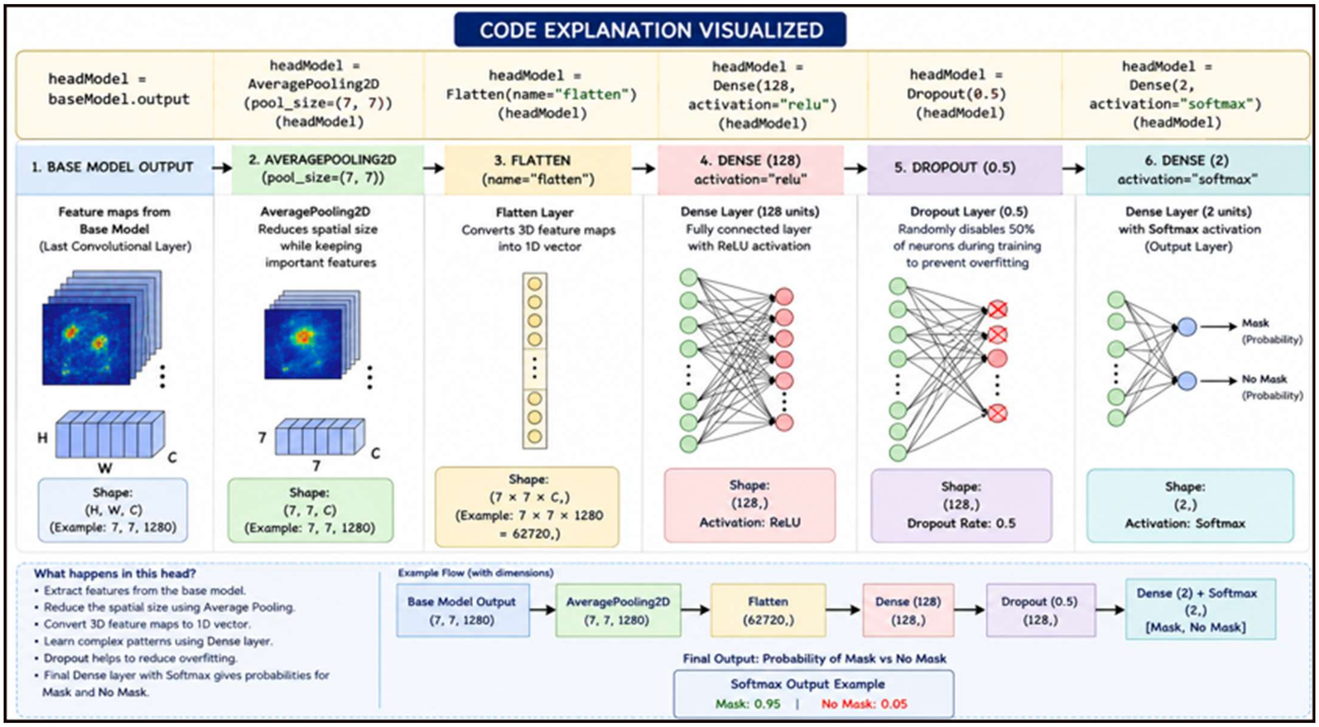


Load MobileNetV2(Transfer Learning)

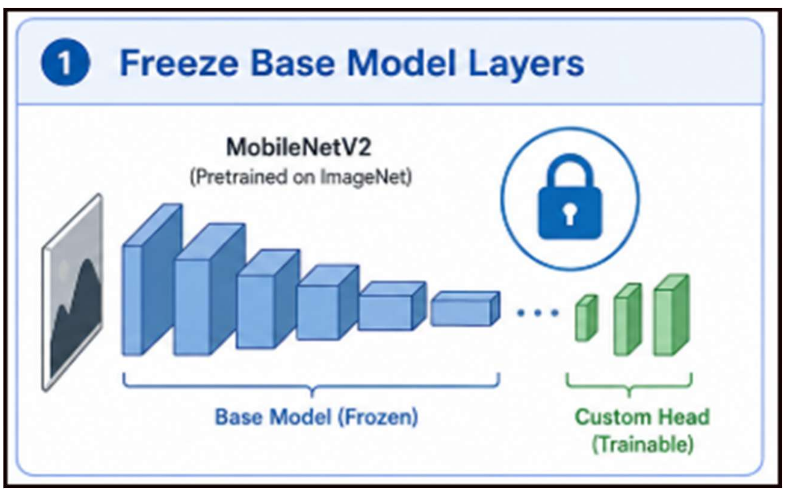




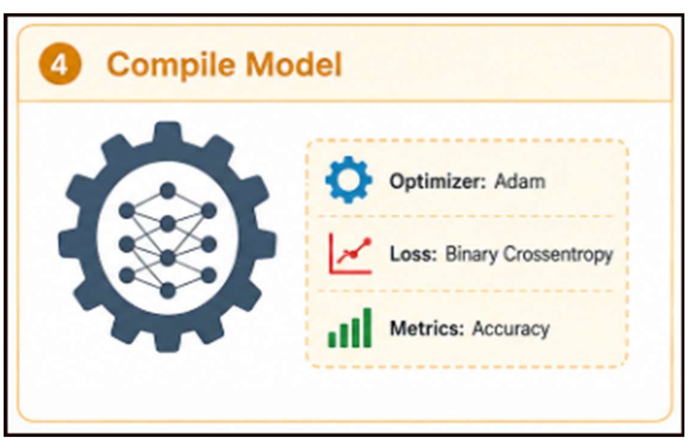
Add Custom Head Layers



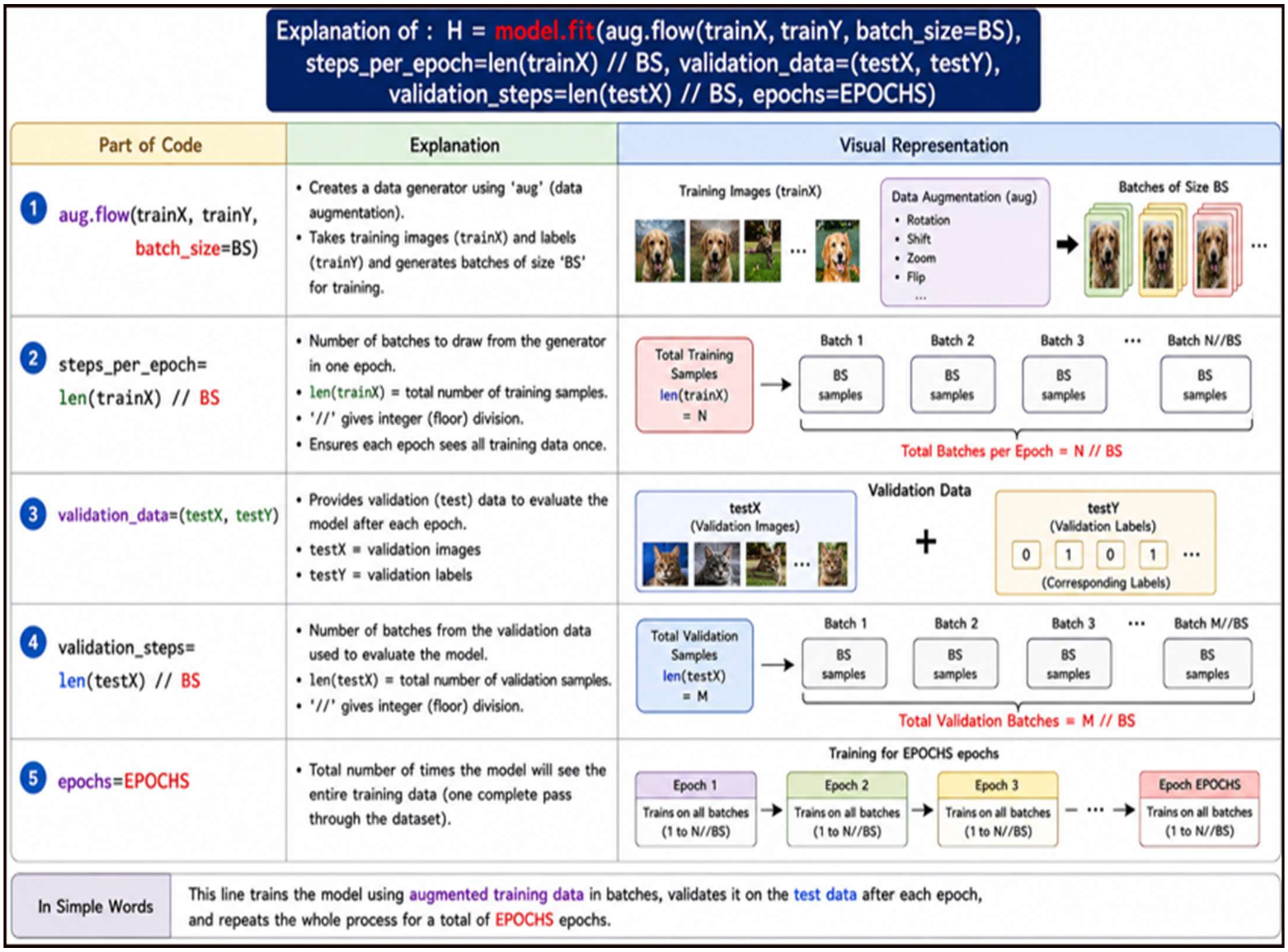
Freeze Base Model Layers



Compile Model



Train Model (20 Epochs)

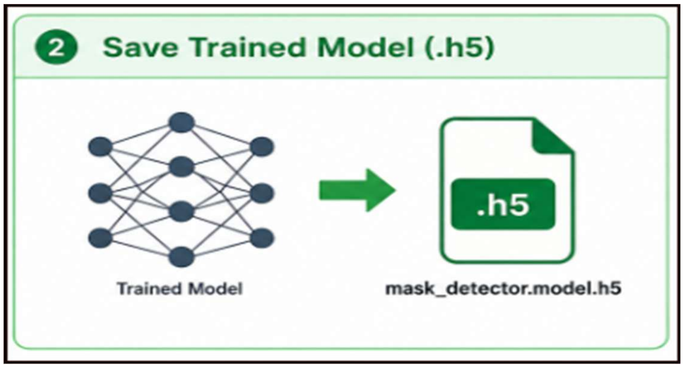


Predict on Test Set

Evaluate Performance

	precision	recall	f1-score	support
0	0.77	0.86	0.81	37584
1	0.84	0.75	0.79	37577
accuracy			0.80	75161
macro avg	0.81	0.80	0.80	75161
weighted avg	0.81	0.80	0.80	75161

Save Trained Model (.h5)



Plot Accuracy & Loss Graph: